

Shubham Singh

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Software engineer with 4 years building **distributed workflow, compliance, and AI application infrastructure** on AWS. Core contributor to control-launch and orchestration platforms supporting 2,000+ annual compliance launches, built PySpark pipelines processing 30M+ records per month, and shipped citation-grounded internal AI tooling across 5 partner teams. Open-source contributor to Giskard.

EXPERIENCE

Amazon

Seattle, WA · Aug 2022 – Present

SDE II (Oct 2025 – Present) · SDE I (Aug 2022 – Oct 2025)

Control Launch & Workflow Platforms · React, TypeScript, Java, Step Functions, DynamoDB

- Drove core backend and UI architecture for an internal control-launch platform, aligning 3 partner teams across 3 time zones and integrating with multiple upstream and downstream systems; the platform cut control launch time from **20+ weeks to days** and grew launch volume 233% year-over-year.
- Designed and solely authored the HLD for a stateful workflow orchestrator replacing brittle Lambda chains with per-marketplace parallel execution, async task-token callbacks, conditional-write locking, and bounded compensation; isolated failures across marketplaces and cut workflow failure rate from **8% to under 1%**.

Data Infrastructure & Operational Leadership · PySpark, AWS, On-call

- Designed a PySpark compliance pipeline processing **30M+ monthly records** with schema versioning, mandatory-field validation, dependent-field validation, and error isolation; reduced incident resolution time by 70%.
- Led a cross-org regulatory schema migration spanning 27 data streams across ~30 teams under a **one-week deadline**; delivered all changes on time and prevented compliance exposure.
- Owned production operations for 5 services, led RCAs for high-severity incidents, authored runbooks, and mentored a new-grad SDE from onboarding to independent on-call readiness.

MaverickIQ: Agentic Q&A Platform · Bedrock, AgentCore, OpenSearch

- Built and shipped to production a citation-grounded internal Q&A platform with hybrid BM25 and kNN retrieval, secure tool calling, tenant isolation, and audit logging; deflects **~90%** of recurring policy and configuration questions that previously required an engineer, across 5 partner teams.
- Engineered a fail-closed grounding policy using Bedrock Guardrails and automated-reasoning checks, achieving **99% verification accuracy** on the automated-reasoning evaluation, and cut vector-storage cost by two orders of magnitude by adopting S3 Vectors.

PROJECTS & OPEN SOURCE

- **Agent Replay** · React, TypeScript, Canvas, Web Workers · 2026
Browser-based time-travel debugger for LLM agent runs; renders **50k+ spans at 60fps** using structure-of-arrays geometry, viewport culling, and LOD aggregation.
- **Giskard OSS Contribution (Merged)** · Python, JSON Schema, LLM Evaluation · 2026
Shipped JsonValid with optional JSON Schema validation into Giskard's LLM evaluation library after maintainer review.

SKILLS

AI & Retrieval: RAG, Hybrid Search, BM25, kNN Retrieval, Citation Grounding, Tool Calling, Guardrails, Evaluation Workflows

Languages: Python, TypeScript, Java, SQL

Frontend: React, Redux, React Hook Form, Component Architecture, Design Systems

Cloud & Platform: AWS Bedrock, Step Functions, Lambda, DynamoDB, OpenSearch, S3, SQS/SNS, EventBridge, CDK, Docker

Backend & Data: PySpark, ETL, FastAPI, GraphQL, REST APIs, Postgres, Redis, OpenTelemetry, Event-Driven Systems

EDUCATION

Indiana University Bloomington: M.S. in Data Science

Jan 2021 – Jul 2022